

Designing Effective Output

CSE 4407

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Output: The 'Face' of Your System

- What
 - Information delivered to users by the system
 - **Channels:** Intranets, Extranets, Web, Mobile Apps, etc.
- Forms
 - **Soft Copy:** Screen displays (desktop, tablet, smartphone), video, audio, social media
 - **Hard Copy:** Printed reports
- **Key Idea:** Users often judge the *entire system's* merit based on its output

The Goal: Creating Useful Output

- **Why:** Useful output drives system adoption and user satisfaction
- **Analyst's Role:** Collaborate *closely* with users
- **Focus:** Six fundamental objectives guide the design of effective output
 - Serve the intended purpose
 - Fit the user
 - Deliver the appropriate quantity
 - Ensure output is where it is needed
 - Provide output on time
 - Choose the right output method

Designing Output: Purpose, Fit, and Quantity

- Serve the intended purpose
 - Align output directly with user/organization goals
 - If output is not functional or purposeful, do not create it
- Fit the user
 - Tailor output to the user's tasks, role, and preferences
 - Personalization/customization is key, especially for interactive systems
 - Consider user needs identified via interviews, observation, prototypes
- Deliver appropriate quantity
 - Provide what's needed to complete work, *but...*
 - Avoid "Information Overload"
 - Consider showing subsets initially, with options to drill down or link for more details.

Delivering Output: Time, Location, and Method

- Ensure output is where it is needed
 - Distribution is crucial, even with digital output
 - Output has no value if it doesn't reach the right decision-maker
- Provide output on time
 - Timeliness is critical for decision-making → Late information = Useless information
 - Consider required frequency (Daily, weekly, monthly, real-time, by exception?)
 - Web-based output can improve timely access
- Choose the right output method
 - Screen, print, audio, video, file export? → Consider cost, accessibility, flexibility, durability, distribution, and impact trade-offs
 - Choice requires careful consideration

Content Meets Method

- **Key:** The *what* (content) influences the *how* (method)
 - Function dictates form
 - Intended purpose guides the choice of output delivery
- **Think Broadly:** Any useful information generated by the system is output
- **Categories**
 - **External Output:** Goes *outside* the business
 - **Example:** Public website content, customer invoices, shareholder reports
 - **Characteristics:** May need instructions, often uses official branding (logo, color), placed on preprinted forms or designed web templates
 - **Trend:** Encouraging electronic versions (online bills) for Green IT and efficiency
 - **Internal Output:** Stays *within* the business
 - **Example:** Intranet pages, summary reports for managers, detailed operational logs, exception reports
 - **Types:** Summaries, detailed, historical, exception-based

Output Technologies Overview

Modality	Example	Advantages	Disadvantages
Printer	Standard, specialized, 3D	Affordable, flexible, high volume, reliable	Needs operator, compatibility issues, supply costs, slow, less green
Display	Monitor, integrated displays, touchscreen	Interactive, real-time, quiet, good for frequent/brief info, uses PC power	Needs space/cabling, save screen views
Audio	Loudspeakers, headphones, earbuds	Good for individuals, transient messages, hands-free use, wide distribution	Needs earbuds for privacy/focus, limited application
Mobile Devices	Smartphones	Portable, interactive, zoomable	Small screen, confusing icons/buttons, easily lost/stolen
Electronic	Web pages, email, RSS feeds, blogs, APIs	Reduces paper, easily updated, broadcast possible, interactive	Formatting limits, context loss, maintenance

Choosing the Right Technology: Key Factors

- **Balancing Act:** Selecting the best tech involves considering many interrelated factors
- **Key Questions**
 - Who will use it?
 - How many people need it?
 - Where is it needed?
 - What is the purpose?
 - What speed is required?
 - How frequently accessed?
 - How long to store it?
 - What regulations apply?
 - What are the costs?
 - What environmental needs?

Understanding Output Users and Quality Needs

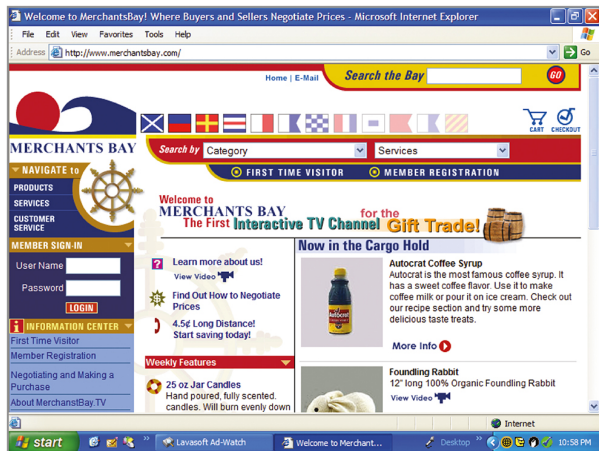
- **Who Will Use the Output?**

- **User Profile**

- Job role, technical skill, location, mobility
 - E.g., mobile manager needs portable output vs. deskbound dispatcher needs screen output

- **Internal vs. External:**

Different needs, different presentation



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Output Volume and Distribution Considerations

- **How Many People Need It?**

- Large audience

- Web documents (easily accessible, scalable) with print options
 - Consider broadcast methods like email lists

- Small audience: Screen display, maybe even audio

- **Where Is Output Needed?**

- Local use? → Intranet pages or local printing
 - Distributed/Remote users?
 - Web or Extranets
 - Electronic distribution
 - Regulations? → Physical copy

Purpose and Timeliness of Output

- **What is the Purpose?**

- Attract investors? → Polished Annual Report (print and web)
- Document history? → Video/Multimedia
- Provide instant update? → Screen crawls, dynamic web content, audio alerts
- Enable user analysis? → Output with embedded tools
- Support Org Processes? → Tracking dashboards, schedules, monitoring alerts

- **What Speed is Needed?**

- Operational Level
 - Needs rapid output for immediate action
 - Online screens are vital
- Strategic Level
 - Needs periodic output for analysis, forecasting, trends
 - Timed reports

Access, Storage, and Regulatory Requirements

- **How Frequently Accessed?**

- High frequency
 - Web/Online systems are ideal
 - Easy access, avoids wear-and-tear on paper
- Low frequency: Archiving (digital, cloud-based)

- **How Long to Store It?**

- Paper degrades → Digital lasts longer, but requires accessible technology
- Driven by government regulations, internal policies, historical value
- Cloud storage offers solutions for long-term digital retention

- **What Special Regulations Apply?**

- Government rules (local, state, federal) influence
 - Output content
 - Storage requirements
 - Distribution rules
- Industry-specific rules

Cost and Environmental Considerations

- **What are the Costs?**

- **Initial Costs:** Hardware purchase/lease (printers, screens), software licenses, development time, web hosting setup
- **Ongoing Costs:** Supplies (paper, ink/toner), maintenance contracts, electricity, website updates, hosting fees, staff time
- **Key:** Look beyond the purchase price!

- **Human and Environmental Requirements?**

- **Human:** Accessibility, ease of use, training needs
- **Physical Environment:** Space for equipment, cabling, Wi-Fi coverage, temperature/humidity control, noise considerations
- **Green IT/Sustainability:** Minimize paper, encourage digital workflows, energy efficiency of devices

Output Isn't Neutral: Understanding Bias

- **Key Concept:** Output actively shapes user perception and decision-making
- **Analyst's Impact:** Choice of data, ordering, wording steers interpretation
- **Bias is Everywhere:** Every report or chart reflects some perspective
- **Goal:** Not to eliminate bias entirely (impossible!), but to:
 - Remove misleading slants
 - Flag remaining biases for users

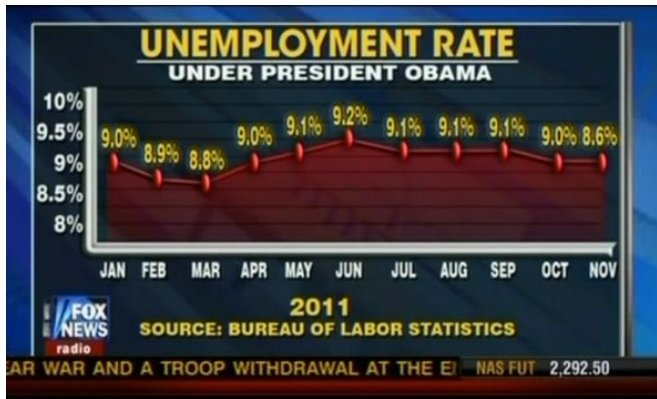
Sources of Unintentional Output Bias

- Common Culprits

- **Sorting:** The order in which information is presented
 - **Primacy Effect:** First items grab attention
 - **Alphabetic Sort:** Emphasizes items starting with A, B, ... → 'Aardvark Printers' get undue attention over 'Zephyr Supplies'
 - **Other Sort:** Chronological or Cost-based sorting can also introduce bias depending on context → SABRE and Apollo reservations systems in 1980s
- **Setting Limits:** Defining thresholds, especially for exception reports
 - **Limits Too Low:** Generates too many "exception," making problems seem bigger or more frequent than they are
 - **Limits Too High:** Real issues might be missed
- **Choice of Graphics:** How data is visualized (type, scale, color, size)

Bias through Graphics

- Visualizations are powerful ...and powerfully misleading if not done carefully
- Bias can come from graph type, size, color choice, and especially scale



Designing for Fairness: Minimizing Output Bias

- **Self-Check Assumption:** Use a bias-review checklist before finalizing design
- **Involve Real Users**
 - Prototype interactions; gather feedback
 - Test across devices, browsers, screen sizes
- **Educate and Warn**
 - Include brief notes on possible limits or biases
 - Offer “info” pop-ups explaining data scope
- **Build Flexibility:** Let users adjust sort order, filters, thresholds
- **Promote “Reality Tests”:** Encourage comparison of multiple reports or data sources

Designing Printed Output: Types and Sources

- **Foundation:** The Data Dictionary → Defines data element names, required field lengths, formats, etc.
- **Types of Printed Reports**
 - **Detailed Reports:** Prints info for **every record on the relevant file**
 - **Use Cases:** Customer mailings, product catalogs, student grade lists
 - **Note:** Many detailed reporting needs are now met by on-screen inquiries
 - **Exception Reports:** Prints info *only* for **records meeting specific criteria** (the “exceptions”)
 - **Use Cases:** Identifying items needing reorder, students on the Dean’s list, accounts significantly overdue
 - **Critical Link:** Requires careful thought about setting limits
 - Primarily used for **operational control**
 - **Summary Reports:** Prints **one line representing a group of records** (e.g., totals, averages)
 - **Use Cases:** Analyzing sales trends, department performance overviews
 - Primarily used for **management decision-making**

Don't Forget Print!

- Reality Check: Users will print web pages
- Design for Printability
 - Offer downloadable PDFs of key documents
 - Test direct printing in Chrome, Firefox, Edge, Safari, etc.
 - Apply `@media print` CSS to hide menus, tweak fonts/colors
- Goal: Clean, readable, professional output

Designing Output for Displays

- Difference from Print
 - Not permanent like paper
 - Can be personalized more easily
 - Available on demand
 - Tied to a device
 - Interactive
- Key Considerations
 - Clear navigation instructions
 - Access control

Guidelines for Display Design

- Core Guidelines

- Keep it Simple

- Avoid clutter
 - Show only what's needed
 - Make info easy to scan

- Stay Consistent: Use uniform layouts, terms, and navigation

- Facilitate Movement

- Enable easy navigation
 - Allow drill-down and backtracking

- Make It Visually Appealing

- Use whitespace, alignment, and clear fonts
 - Pick readable colors

- Get User Feedback

- Use prototypes
 - Iterate based on real input

Example: Order Status Display

- Simple Layout
 - Shows key status info only
 - No visual clutter
- Consistent (Implied): Sample style used across status screens
- Supports Navigation: Clear instructions
- Visually Clear
 - Bold heading
 - Tabular view for easy reading
 - Includes context for user

New Zoo Order Status			
Retailer	Order #	Order Date	Order Status
Animals Unlimited	933401	09/05/2022	Shipped On 09/29
	934567	09/11/2022	Shipped On 09/21
	934613	09/13/2022	Shipped On 09/21
	934691	09/14/2022	Shipped On 09/21
Bear Bizarre	933603	09/02/2022	Partially Shipped
	933668	09/08/2022	Scheduled For 10/03
	934552	09/18/2022	Scheduled For 10/03
	934683	09/18/2022	Shipped On 09/28
Cuddles Co.	933414	09/12/2022	Shipped On 09/18
	933422	09/14/2022	Shipped On 09/21
	934339	09/16/2022	Shipped On 09/26
	934387	09/18/2022	Shipped On 09/21
Stuffed Stuff	934476	09/25/2022	Backordered
	934341	09/14/2022	Shipped On 09/26
	934591	09/18/2022	Partially Shipped
	934633	09/26/2022	Backordered
	934664	09/29/2022	Partially Shipped

Press any key to see the rest of the list; ESC to end; ? for help.

For more detail place cursor over the order number and hit the Enter key.

Example: Facilitating Movement

- Facilitates Movement: Users can drill down from summary to detail
- Keeps Initial Screen Simple
 - Only essential info shown at first
 - Prevents screen clutter
- Consistent Instructions
 - Same help/navigation cues at bottom
 - Builds user familiarity
- Use Control
 - Users decide when and if they want more detail
 - No forced navigation

Order #	Retailer		Order Date	Order Status	
933603	Bear Bizarre 1001 Karhu Lane Bern, Virginia 22024		09/02/2022	Partially Shipped	
Units	Pkg	Description	Price	Amount	Detailed Status
12	Each	Floppy Bears	20.00	240.00	Backordered Due 10/15
6	Each	Growlers	25.00	150.00	Backordered Due 10/15
2	Each	Special Edition	70.00	140.00	Shipped 09/02
1	Box	Celebrity Mix	150.00	150.00	Shipped 09/02
12	Each	Polar Bears	10.00	120.00	Backordered Due 10/30
				800.00	
Contact	Account Balance		Credit Rating	Last Order	Shipped
Ursula Major 703-484-2327	0.00		Excellent	08/21/2022	On Time

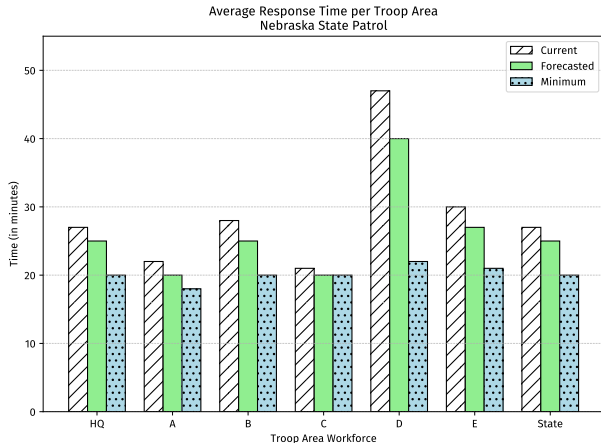
Press any key to see the rest of the list; ESC to end; ? for help.

Using Graphical Output on Displays

- Why
 - Spot trends and patterns faster than with tables
 - Visually compare values at a glance
- Key Considerations
 - Work with users: Choose the *right* graph for message
 - Keep it clear: Must be easy to read and accurate
 - Expose assumptions: Explain scales or axes to avoid misinterpretation
- Design Factors
 - Purpose: Inform, compare, show trends
 - Data Type: Time series, categories, distributions
 - Audience: Experts vs. novices
 - Perception: Graph types influence how users interpret data

Example: Nebraska Patrol DSS Graph

- **Intelligence:** Flags problem areas → Troops D and E have high response times
- **Design:** Compare current, forecast, and minimum times → See potential impact
- **Choice:** Support smart resource allocation



Dashboards: Information at a Glance

- What

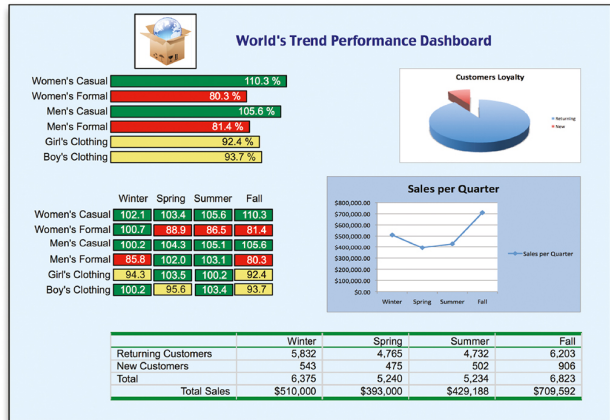
- Like a car dashboard → Multiple indicators, one screen
- Used by execs to monitor key metrics (KPIs) quickly
- Shows
 - Graphs and speedometer-like visuals
 - Warnings/alerts like engine lights
 - Key text/figures like odometers

- Goal

- Give a clear snapshot of performance
- Support fast decisions with digestible data
- But... only if well-designed

Example: World's Trend Performance Dashboard

- All key views on one screen
- Helps manager **assess performance from many angles**
- Tools like Tableau make this possible



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Dashboard Design Rules

- Provide Context
 - Don't just show a number
 - Show it relative to goals, history, or benchmarks
 - Example: Is \$851k sales good or bad?
- Summarize Wisely
 - Avoid excessive detail
 - Use summaries where appropriate
 - Example: \$851,235.32 vs. \$851k
- Choose the Right Metrics: Variances vs. Raw numbers
- Stay Fair
 - Avoid distortions
 - Example: Misleading scales
- Correct Graph Style: Line for trends, bar for comparison

Dashboard Design Rules (Cont.)

- **Good Layout:** Clear labels, smart color user
- **Limit Chart Types:** Too much variety = Visual noise
- **Highlight Wisely:** Bold/red for key items
- **Group Logically:** Related metrics = Side by side
- **Keep it Clean:** No flashy logos or background art
- **One Screen:** Avoid scrolling or tabs
- **Be Flexible:** Gather feedback, iterate often

Infographics: Visual Story telling

- What
 - A data story told visually
 - Combines numbers, charts, icons, and text
- When to Use
 - Engages audiences more than plain tables
 - Makes data fast and easy to grasp
 - Ideal for reports, marketing, or internal updates
- Tools: Web apps like Canva, Piktochart, etc.
- Example
 - Mixes standard and custom visuals
 - Clear message
 - Clean design with a limited palette



Infographics Design Tips

- Start Strong: Use a newsworthy headline
- Tell a Story: Arrange visuals with purpose
- Mix Elements: Charts, icons, maybe cartoons? → Skip busy photos
- Keep It Brief: Visuals carry the message
- Limit Colors: Stick to 2-4 coordinated shades
- Call to Action: Add contact info if relevant
- Design to Share: Perfect for social media

Designing a Website: Beyond a Single Page

- From single “pages” to full “websites”
- First impression = landing page
- Web → An uncontrolled environment: Different
 - Browsers
 - Screen sizes/resolutions
 - Devices (PCs, tablets, phones)
 - User settings (fonts, JS)
- **Big Challenge:** Design for **variability**
- Builds on display design, but adds web-specific twists

Designing for All Screens: Responsive and Adaptive

- **Responsive Web Design (RWD)**
 - One site, works on all devices
 - Uses **fluid grids** (percent-based layout)
 - Content **reflows with screen**
 - Ads may not be flexible → Use **Ajax or alternate landing pages**
- **Adaptive Web Design (AWD)**
 - **Detects device, loads best layout**
 - Uses **fixed grids for desktop, tablet, and mobile**
 - **Offers more control** across device types

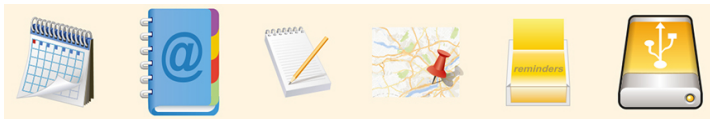
Building Websites: Tools and Editors

- Cloud-Based WYSIWYG: Weebly, Wix
 - Drag and drop, no coding
 - Fast, easy, freemium model
 - Downside: Host lock-in, limited free features
- Stand-Alone Software: Dreamweaver
 - Full features, no ads
 - Own your files → Host anywhere
 - Costs more, steeper learning curve

Design Trends: Skeuomorphic vs. Flat

- **Skeuomorphic Design**

- Mimics real-world objects
- Uses shadows, textures, fake realism
- **Became outdated**



- **Flat Design**

- Clean, minimal, 2D
- No shadows, simple shapes
- **Focuses on clarity and function**



- **Trends Evolve: Flats may fade; 3D realism may return**

Preparing for Web Design: Tools, Research, and Inspiration

- Use Design Tools
 - Stand-alone (Dreamweaver) or cloud (Wix)
 - Faster and richer than hand-coding
- Study Great Sites
 - What works? What's engaging?
 - Structure, features, flow
- Learn from Pros: Critique design, interactivity, colors, metaphors
- Evaluate with Tools: Use forms/checklists to rate sites

Bringing Ideas to Life: Planning, Prototyping, and Best Practices

- Plan the Site First
 - Storyboards: Screen flow like movie scenes
 - Wireframes: Basic layout → Boxes, no style
 - Mockups: Styled, realistic screens before coding
- Use Templates: Set layouts → Faster + Consistent design
- Use CSS
 - Separates content from style
 - One style change updates many pages
- Be Cautious with Plug-Ins
 - Flash, extensions → Not always supported
 - Stick to broadly compatible tech

Specific Website Design Guidelines

- **Structure:** Logical organization and planning.
- **Content:** Valuable and engaging information/features.
- **Text:** Clear, readable, well-formatted writing.
- **Graphics:** Appropriate use of images.
- **Presentation Style:** Overall look, feel, and layout.
- **Navigation:** How users move through the site.
- **Promotion:** Making the site discoverable.

Design Details: Structure and Content

- Structure
 - Define goals and page purposes
 - Plan site with mapping tools
 - Avoid broken links
- Content
 - Provide Value: Tips, tools, interactivity
 - Boost “stickiness” (time-on-site)
 - Use helpful metaphors (sparingly!)
 - Add FAQ page
 - Include widgets
- Examples: Polygon, Arngren Electronics

Design Details: Text and Visuals

- Text

- Clear page titles
- Strong opening sentence
- Simple, scannable writing
- Keep content fresh

- Graphics

- Use PNG/JPEG/GIF wisely
- Simple backgrounds
- Small file sizes = Faster load
- Reuse elements for caching
- Add tooltips for images
- Test on multiple devices/screens

- Examples: McMaster-Carr, Berkshire Hathaway, Yale School of Art

Design Details: Consistent and Fast Experience

- Fast home page (<3s, <100KB)
- Simple menus, clear options
- Limit heavy graphics
- Big, colorful headings
- Use image maps for creative links
- CSS for control and consistency
- Use <div>s, not tables, for layout
- Use JavaScript for interactivity, not annoyance
- Examples: [FormaFantasma](#), [James Bond Museum](#)

Design Details: Smart Navigation

- Consistent Nav Bar + Link to home on every page
- Clear link label
- Follow standard patterns
- Use breadcrumbs (Home > Section > Page)
- Avoid confusing jargon or menu names [1]
- 3-Click Rule
 - Not strict → Focus on user success
 - Better to go deeper if it helps clarity [2]
- Mega Menus
 - Big, hoverable menus → Lots of options, no scrolling
 - Use icons, categories, layout
 - Make parent link clickable for accessibility
- Examples: Amazon, CogniFi

Design Details: Promotion

- Submit your site to search engines
- Add keywords in `<meta>` tags
- Use email carefully → Avoid spam
- Make site worth bookmarking
- Add a favicon for recognition
- Link out wisely → Give users a way back

Integrating Collaboration: Web 2.0 Technologies

- What
 - Static pages → Interactive participation
 - Emphasizes user content and collaboration
- Key Technologies
 - Blogs: Article with reader comments
 - Wikis: Collaborative editing
 - Social Media Links: Facebook, Twitter, LinkedIn, YouTube
 - Tagging/Bookmarking: User-assigned keywords for content delivery
- Applications
 - External: Engage customers and partners
 - Internal: Boost employee collaboration

Strategic Goals for Web 2.0 Integration

- External Goals

- Consistent branding across platforms
- Collect feedback and monitor opinions
- Build online user community
- Improve customer support and engagement

- Internal Goals

- Foster trust and relationships
- Share knowledge, boost innovation
- Tag resources for easy access
- Support company culture

- **Decision Tip:** Use tools only if aligned with user needs (found during analysis/prototyping)

Analyst Considerations for Web 2.0

- **Align Objectives:** Balance company goals vs. user expectations
- **Be Customer's Voice:** Advocate for user needs in external designs
- **Design with Conventions**
 - Place icons where users expect (header/footer)
 - Use standard, familiar symbols
 - Follow layout norms for usability
- **Revise and Update**
 - Web tools and trends evolve fast
 - Plan for regular updates
- **Integrate with Branding**
 - Match tools with company style and voice
 - Keep look and tone consistent

Social Media Icon Conventions

Name of Icon Set	Designer or Supplier	Facebook	LinkedIn	RSS Feed	Twitter
Social Networks Pro Icons	Artbees				
Circular Social Media	BlogPerfume.com				
Free Social Media Icon Set	Elegant Themes				
Amazing 3-D Social Icons	pinkmoustache.net				
Vintage Icons for Bloggers	Nikola Lazarevic				

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Courtesy: Facebook, LinkedIn, and Twitter

Introduction to App Design

- What
 - Evolved Term: Program → Application → App
 - Popularized by Apple; now standard
- Design Process
 - Begins with brainstorming
 - UI Design Tools: Figma, Adobe XD, Sketch, Framer, Marvel App, Proto.io, InVision, etc.
 - Developer-driven motivation; informal start
- Focus: Small screens (phones/tablets); principles apply across platforms

Getting Started: Planning and Originality

- Developer Account
 - Apple: Strict approval (“walled garden”)
 - Android: Open platform
- Development Process
 - Prototyping preferred → Agile
 - Release early, iterate fast
 - Visibility boost via frequent updates
- Be Original
 - Don't copy
 - Avoid trademarked names

Monetization and Branding Strategy

- Pricing Strategy
 - Low-Cost: \$0.99-\$1.99 → High volume
 - Premium: \$19.95+ → Niche/Pro users
 - Freemium: Free + IAPs
 - Free: Portfolio/promotional
 - Ad-Supported: Requires UI space
 - Promos: Temporary visibility boosts
- Follow Platform Rules: Apple's Human Interface Guidelines
- Design the Icon
 - Must be memorable, scalable
 - Preview tools like App Cooker
- Choose a Name: Max ~12 characters → Avoid truncation

Designing for Devices and Prototyping

- Device Screen Sizes and Icon Scaling
 - Challenge: Many screen sizes even within OS
 - Use Vector Graphics
 - Design large
 - Scale down without quality loss
 - Great for print too
- Design Output Screens
 - Use drag-and-drop UI widgets
 - Follow screen design principles
- Handle Orientation
 - Support both Portrait and Landscape
 - Offer enhanced features in landscape
- Share Prototypes: Tools like AppTester for previews

Building Functionality and Going to Market

- Design App Logic: Use process specs/decision trees
- Design Navigation: Plan screen flow/storyboards
- Gestures
 - Swipes, Pinches, Taps, Long presses
 - Provide feedback, consider opt-out
- IP Protection
 - Trademark icon/logo
 - Copyright code/design
 - Write a custom EULA
- App Store Page
 - Clear description
 - What's new
 - Screenshots to show best features

References I

- [1] R. Budiu, “Information scent: How users decide where to go next,” (2020), [Online]. Available: <https://www.nngroup.com/articles/information-scent/> (cit. on pp. 121–125).
- [2] J. Nielsen and A. Li, “Mega menus work well for site navigation,” (2017), [Online]. Available: <https://www.nngroup.com/articles/mega-menus-work-well/> (cit. on pp. 121–125).